

heap block b

active allocation

header.lock = k
new pointer carries k

delete

free, still mapped

header.lock = SENTINEL
old pointer still carries k

reuse b

new active allocation

header.lock = k' (k' != k)
old pointer fails live; new pointer passes

stable slot l

frame entry

slot[l] = k
escaped pointer carries k

return

inactive frame

slot[l] = SENTINEL
escaped pointer fails live

slot reuse

fresh frame lifetime

slot[l] = k' (k' != k)
old pointer remains invalid